

Mahesh Panta

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Summary

Computer Engineering undergraduate at Kathmandu University with hands-on experience in machine learning, computer vision, and 3D reconstruction research. Currently a Research Trainee at KU's HI Lab, working on depth estimation and 3D scene reconstruction. Proficient in building deep learning pipelines (TensorFlow, PyTorch), training object detection models (YOLOv8), and deploying ML-powered APIs with FastAPI and PostgreSQL. Passionate about applying computer vision, reinforcement learning, and full-stack AI to real-world problems.

Education

Kathmandu University

Bachelor of Engineering in Computer Engineering

Dhulikhel, Nepal

Present (3rd Year)

Khwopa Secondary School

NEB Plus Two (Science with Biology) | GPA: 3.7

Nepal

Graduated

Experience

Research Trainee - Computer Vision & 3D Reconstruction

Mar 2026 – Present

HI Lab, Kathmandu University

Dhulikhel, Nepal

- Conducting research in computer vision and 3D scene reconstruction as part of KU's Human-Intelligence Lab.
- Working on depth estimation, point cloud generation, and computer vision pipelines using Python and OpenCV.
- Collaborating with lab members on applied CV research targeting real-world spatial understanding problems.

Technical Skills

Programming Languages: Python, C, C++

Machine Learning & AI: Supervised & Unsupervised Learning, CNNs, Transfer Learning, Data Augmentation, Model Optimization, Recommender Systems

Computer Vision: YOLOv8, OpenCV, 3D Reconstruction, Depth Estimation, Structure-from-Motion, NeRF, MobileNet, Metric Learning

Reinforcement Learning: Q-Learning, Model-Based Agents, Reflex Agents, Environment Simulation, Contextual Bandits (UCB)

Deep Learning Frameworks: TensorFlow, Keras, PyTorch (fundamentals)

Data Science: NumPy, Pandas, Matplotlib, Scikit-learn

Backend & APIs: FastAPI, Flask, REST API Design

Databases: PostgreSQL, SQLite, SQLAlchemy

Mobile & Frontend: Flutter (Dart), Kivy, React, HTML, JavaScript (Beginner)

Game Development: C++ with Raylib, RayGUI

Networking: ENet (Reliable UDP), REST APIs

Tools & OS: Git, CMake, Ubuntu, Jupyter Notebook

Languages: English, Nepali, Hindi, Japanese

Projects

ChitraDartaa Civic Issue Reporting System

GitHub

Flutter, FastAPI, PostgreSQL, YOLOv8, TensorFlow

- Built a full-stack civic issue reporting platform enabling citizens to submit geo-tagged image reports (potholes, garbage, road damage) with GPS location and resolution status tracking.
- Trained YOLOv8 object detection and segmentation models to automatically classify issue types from uploaded images.
- Developed CNN-based classifiers (road-issue vs non-issue, potholes vs garbage) with TensorFlow/Keras, applying data preprocessing and augmentation pipelines.

- Designed a FastAPI REST backend with PostgreSQL (via SQLAlchemy) for persistent storage of reports, user data, and resolution tracking.
- Built a cross-platform Flutter mobile frontend with interactive map integration, push notifications, and an admin dashboard for authorities.

Clothing Visual Search & Matching System

GitHub

PyTorch, TensorFlow, FastAPI, React, PostgreSQL

- Co-developed a full-stack visual search system that scans an uploaded clothing image and retrieves the closest matching item from a labeled wardrobe dataset spanning 25 categories.
- Benchmarked ResNet, a ResNet encoder with CNN decoder, MobileNet, and YOLO architectures for feature extraction, training with metric learning (triplet loss, online batch-hard mining) to build a similarity embedding space.
- Applied a Linear UCB (contextual bandit) strategy within the matching pipeline to guide result ranking.
- Auto-generated YOLO-format labels with LangSAM and used class-balanced oversampling to improve accuracy on visually confusable categories.
- Built a FastAPI + PostgreSQL backend with JWT authentication and a React frontend for uploading scans and browsing matches.

Forest Conservation System

GitHub

Python, Computer Vision, Machine Learning

- Developed an AI-powered forest monitoring system for detecting deforestation and environmental threats from image and remote sensing data.
- Applied computer vision techniques for change detection and classification of forest cover using deep learning models.
- Integrated ML pipelines for automated environmental risk assessment.

Game Vault: Mini Game Arcade with AI Agents

GitHub

C++, Raylib, ENet

- Co-developed a multi-game arcade (Pong, Snake, Space Invaders) with real-time multiplayer via ENet reliable UDP networking.
- Implemented RL-inspired AI agents: a Simple Reflex Agent for reactive play and a Model-Based Agent for stateful environment tracking.
- Managed cross-platform builds and dependency resolution using CMake.

YummY-Chef Chatbot

GitHub

Python, Kivy, ChatterBot, SQLite

- Built a mobile-friendly chatbot providing recipe suggestions and cooking assistance using ChatterBot NLP and a Kivy cross-platform UI.
- Implemented persistent data storage and conversation memory with SQLite.

Achievements

- **Locus HACK-A-WEEK Finalist** (Pulchowk LOCUS Hackathon 2026) – Advanced to the finals among top competing teams with the *Forest Conservation System* project. [LinkedIn Post]

Certifications

- **Unsupervised Learning, Recommenders, Reinforcement Learning** – Deep Learning.AI (Apr 2026, Credential: 49W64MVQ5LQ9)
- **Machine Learning Specialization** – Deep Learning.AI (Apr 2026, Credential: JUBLWYJDIHKT)
- **Advanced Learning Algorithms** – Deep Learning.AI (Nov 2025, Credential: BQCH8WWP8VMT)
- **Supervised Machine Learning: Regression and Classification** – Deep Learning.AI (Oct 2025, Credential: U0574LHEIT2T)
- **Introduction to Programming Using Python** – CS50, Harvard University
- **Pandas** – Kaggle